

Empirical Strategy Blueprint

Fertility and Housing Supply

Fertility and Housing Supply Project

February 20, 2026

Purpose

This note ranks candidate empirical designs and provides an implementation blueprint for the top two. The ranking criterion is practical: high identification credibility with feasible data assembly in finite time.

1 Ranking summary

Design	Rank
State middle-housing legalizations and fertility outcomes	1
NIMBY political turnover (close elections) and fertility via supply	2
Family-sized unit composition shocks (shift-share IV)	3
Short-term rental crackdown shocks and fertility	4
Single-city mega-reform with legal interruptions	5

2 Top design 1: Middle-housing legalizations

Research question

Do reforms that legalize middle housing (duplex/triplex/fourplex forms) increase fertility among prime childbearing populations by improving access to family-suitable housing?

Estimand

Dynamic average treatment effect on treated locations (ATT) for age-specific birth rates.

Identification strategy

Staggered event-study difference-in-differences with treatment intensity.

Baseline event-study:

$$BirthRate_{it} = \sum_{k \neq -1} \beta_k 1\{EventTime_{it} = k\} + \alpha_i + \delta_t + X'_{it}\Gamma + \varepsilon_{it}.$$

Intensity extension:

$$BirthRate_{it} = \beta(Post_{it} \times Exposure_i) + \alpha_i + \delta_t + X'_{it}\Gamma + \varepsilon_{it}.$$

Data build

- Natality outcomes by mother's age from public vital-statistics sources (county-year minimum).
- Permits and housing production composition (single-family vs multifamily; where possible, unit-size proxies).
- Price/rent controls and labor-market controls.
- Policy adoption and effective-implementation timing panel.

Main identifying assumptions

- In absence of reform, treated and untreated areas follow parallel fertility trends conditional on fixed effects and controls.
- No confounding policy shocks perfectly collinear with reform timing after controls.
- Exposure intensity captures meaningful heterogeneity in reform bite.

Falsification and robustness

- Pre-trend tests and placebo policy dates.
- Alternative treatment windows (announcement vs implementation).
- Alternative exposure definitions.
- Re-estimation excluding major outliers and migration-heavy areas.

Strengths and risks

Strengths: policy relevance, scalable geography, direct link to housing supply.

Risks: first-stage quantity response may be slow; fertility response may lag reform substantially.

3 Top design 2: NIMBY political turnover design

Research question

Does narrowly electing pro-housing local governments shift housing supply and affordability, and through this channel alter fertility outcomes?

Estimand

Two-step causal object:

1. effect of pro-housing victory on supply/price outcomes;
2. effect of induced supply changes on fertility.

Identification strategy

Close-election RD combined with IV-style mediation.

First-stage equation:

$$Supply_{it} = \pi ProHousingWin_{it} + f(Margin_{it}) + \eta_i + \tau_t + u_{it}.$$

Second-stage equation:

$$BirthRate_{it} = \beta \widehat{Supply}_{it} + f(Margin_{it}) + \eta_i + \tau_t + e_{it}.$$

Data build

- Local election returns with close-race margins.
- Candidate/council platform coding for housing stance.
- Permit and unit-composition outcomes.
- Price/rent outcomes and fertility outcomes mapped to compatible geography.

Main identifying assumptions

- Near the cutoff, winners and losers are quasi-random.
- Platform coding is sufficiently accurate to separate pro-housing from anti-growth outcomes.
- Exclusion for second stage: election-induced fertility impacts operate primarily through supply/price channels (to be argued carefully).

Falsification and robustness

- McCrary-style density and covariate-balance checks at the cutoff.
- Placebo cutoffs and placebo outcomes.
- Alternative bandwidths and polynomial orders.
- Robustness to coding uncertainty (drop ambiguous races).

Strengths and risks

Strengths: direct NIMBY political-economy identification; strong conceptual fit.

Risks: heavy data engineering and potential power constraints from close-race sample size.

4 Why these two are the best portfolio

These top two designs are complementary:

- Design 1 delivers broad external validity and cleaner implementation.
- Design 2 delivers a sharper political-economy test tied directly to your paper’s NIMBY channel.

Running both in parallel gives a “safe” main design and a “high-upside” political design.

5 Operational sequencing

Phase 1 (fast start)

1. Build policy timing panel and core outcome panel.
2. Estimate first-pass event-study for design 1.
3. Verify first stage on permits and composition.

Phase 2 (NIMBY track)

1. Build election and coding dataset.
2. Run close-race diagnostics and first stage.
3. Add second-stage fertility mapping where geography allows.

Bottom line

For a credible and publishable empirical chapter, begin with middle-housing reforms as the baseline design. In parallel, develop the close-election NIMBY design as the distinctive contribution that links your theory directly to political supply restrictions and demographic outcomes.